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Hand-aware graph convolution network for skeleton-based sign language recognition

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ABSTRACT

Skeleton-based sign language recognition (SLR) is a challenging research area mainly due to the fast and complex hand movement. Currently, graph convolution networks (GCNs) have been employed in skeleton-based SLR and achieved remarkable performance. However, existing GCN-based SLR methods suffer from a lack of explicit attention to hand topology which plays an important role in the sign language representation. To address this issue, we propose a novel hand-aware graph convolution network (HA-GCN) to focus on hand topological relationships of skeleton graph. Specifically, a hand-aware graph convolution layer is designed to capture both global body and local hand information, in which two sub-graphs are defined and incorporated to represent hand topology information. In addition, in order to eliminate the over-fitting problem, an adaptive DropGraph is designed in construction of hand-aware graph convolution block to remove the spatial and temporal redundancy in the sign language representation. With the aim to further improve the performance, the joints information, bones, together with their motion information are simultaneously modeled in a multi-stream framework. Extensive experiments on the two open-source datasets, AUTSL and INCLUDE, demonstrate that our proposed algorithm outperforms the state-of-the-art with a significant margin. Our code is available at <https://github.com/snorlaxse/HA-SLR-GCN>.

1. Introduction

Sign language is the primary communication way among millions of deaf people around the world. Sign language, including a series of hand gestures, motions and transitions, conveys semantic meaning. To further facilitate communication between deaf-and-dumb sign language users and healthy people who do not understand sign languages, sign language recognition (SLR) which recognizes signs automatically from given videos, has become a hot research direction.

Sign language modality can be classified into RGB, depth, flow and skeleton data [1]. Among these modalities, skeleton data, which contains the coordinates of the human body keypoints and filters the environmental noise, has strong robustness to the dynamic

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circumstances and complicated backgrounds [2–4]. Recently, with the development of 3D sensors [5] and pose estimation algorithms [6,7], the skeleton data become easily obtained and skeleton-based SLR has been attracted increasing attention.

Early works adopted hand-crafted features and machine learning methods to perform skeleton-based SLR [8–10]. With the development of deep learning methods, SLR researches have been developed rapidly in recent years. Time-series representation learning methods (e.g., RNN, its variant LSTM) [11] are explored for SLR tasks. These methods considered skeletal sequences as individual frames, where temporal and spatial information were processed separately. To jointly extract the spatial-temporal feature, some methods [12,13] performed feature extraction with a backbone of 2D or 3D CNNs over the skeleton sequence. However, the RNN-based and CNN-based methods transformed sampled skeleton joints into gridded 1D vector, which failed to explore the graph structure of human body and the interdependence between skeleton joints.

In recent years, due to its extraordinary ability in modeling non-Euclidean structures, graph convolutional networks (GCNs) have received extensive attention. Some approaches try to adopt GCNs in skeleton-based SLR. The first successful work was the spatio-temporal graph convolution network (ST-GCN) [14], which utilized human joint connections to make spatial and temporal connections across frames to construct 3D topology graph. Jiang et al. [15] adopted an extended variant of the spatial graph convolution called decoupling graph convolution [16] to further boost the capacity of GCN for SLR. However, they have difficulty in modeling the long-range relationships and capturing complex spatial-temporal dependencies. To effectively model cross spatial-temporal correlations, Liu et al. [17] presented a multi-scale spatial-temporal graph convolution (MS-G3D) for SLR. Meanwhile, Al et al. [18] also proposed a graph-based architecture based on separable 3D CNN (S3D) for SLR, and used a multi-head attention mechanism to enhance the spatial context representation of the gestures. However, existing GCN-based solutions did not explore the prior knowledge of hands even if some attention mechanisms are used.

Considering the importance of hand information in SLR, we propose a novel hand-aware graph convolution network (HA-GCN) in an end-to-end manner to impose a focus on hand topological relationships of skeleton graph. As illustrated in Fig. 1, we design a hand-aware graph convolution layer to capture both global body and local hand information. Alongside original upper body information, two sub-graphs representing hand topology information are defined and incorporated in the graph convolution. In addition, an adaptive DropGraph strategy is elaborately designed in the construction of the hand-aware graph convolution block to remove the spatial and temporal redundancy in the sign language representation. That would eliminate the over-fitting problem. Moreover, a multi-stream approach is implemented to further improve the SLR performance, which simultaneously combines various sequence representations of both the joints and bones, together with their motion information.

Overall, the main contributions of this paper can be summarized as follows:

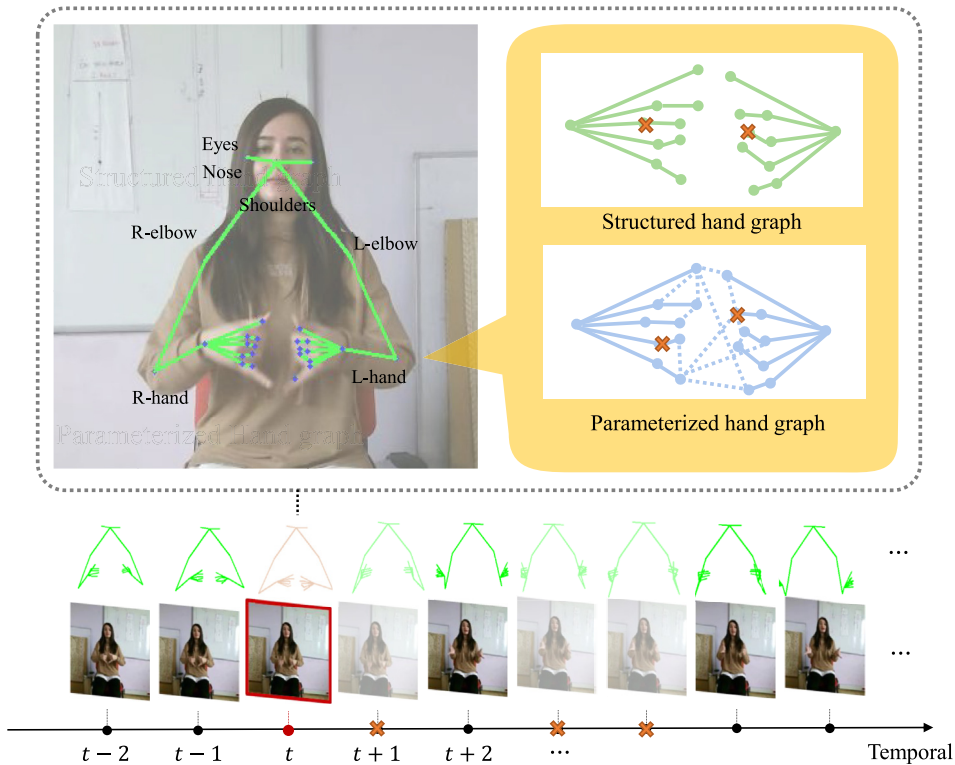


Fig. 1. The illustration of proposed hand-aware GCN for SLR. Two different adaptively learned hand-graphs (structured hand graph and parameterized hand graph) are introduced in the HA-GCN layer to highlight the hand topology. And adaptive DropGraph strategy is designed to discard the redundant features in spatial and temporal skeleton sequence. The red crosses indicate adaptively dropped spatial and temporal information.

- (1) A novel hand-aware graph convolution network (HA-GCN) is proposed to focus on hand topological relationships of skeleton graph, where two different hand-aware sub-graphs are defined and incorporated in our proposed HA-GCN layer to highlight the hand topology. The graph topology in our model can be adaptively learned based on the input data in an end-to-end manner.
- (2) An adaptive DropGraph strategy is elaborately designed to eliminate the over-fitting problem. Both the spatial and temporal DropGraphs are adaptively incorporated through a gating mechanism to discard redundant frames and nodes.
- (3) Extensive experiments are conducted on the two opensource AUTSL and INCLUDE datasets. Experimental results demonstrate that our proposed HA-GCN can adaptively learn the hand topological relationships. The top-1 accuracies of our proposed model for two datasets reach 96.82% and 99.63%, respectively. The SLR performance of our proposed HA-GCN exceeds the state-of-the-art with a significant margin.

2. Related work

In this section, we briefly review the related works from two aspects: skeleton-based sign language recognition (SLR) and graph convolution neural network (GCN).

2.1. Skeleton-based sign language recognition

3D skeleton based methods for SLR has recently attracted a lot of research attention because of its high level representation and robustness to variations of viewpoints, appearances, and surrounding distractions. Early skeleton-based methods usually exploited local features and sequential state transitions to represent temporal dynamics. Wang et al. [19] first extracted the local occupancy patterns from the appearance around skeleton joints, and then processed them with Fourier temporal pyramid (FTP) to obtain temporal structure. Lv et al. [20] extracted local features of individual and partial combinations of joints, and trained hidden Markov models (HMM) to capture the action dynamics. Though HMM can model the temporal evolution of actions, the input sequences have to be segmented and aligned, which is difficult to carry out.

With the development of deep learning methods, SLR research has produced great advancement for feature extraction. Time-series representation learning methods (e.g., RNN, its variant LSTM) [12] are explored for SLR tasks. These methods considered skeletal sequences as individual frame, where temporal and spatial information is processed separately. In order to jointly extract the spatial-temporal texture feature, some methods [11,13] considered a sequence of coordinates of human body keypoints as a pseudo-image and performed with 2D convolution neural networks (CNNs) in the group of frames. For example, PoseConv3D [13] adopted a skeleton-based CNN approach, which relied on a 3D heatmap volume instead of a graph sequence as the base representation of human skeletons. However, these methods, which represented the skeleton sequence as vector sequence or a pseudo-image, ignored the intrinsically kinematic dependencies and topologies of the human body.

In recent years, graph convolution networks (GCNs) have achieved great success in SLR tasks, due to its extraordinary ability in modeling topology structures [16,21,22]. These models are based on spatial-temporal GCN (ST-GCN) architecture [14,23,24], which designed custom graph representation layouts to be appropriate for SLR and effectively model spatio-temporal correlations. Specifically, Amorim et al. [24] directly applied ST-GCN to a new dataset ASLLVD of human skeletons for sign language. Jiang et al. [15] adopted an extended variant of the spatial graph convolution called decoupling graph convolution [16] to further boost the capacity of GCN for SLR. However, a limitation exists in their ability to model long-range relationships and capture complex spatio-temporal dependencies. To this end, MS-G3D [17] is proposed to design a multi-scale spatial-temporal graph convolution with disentangling the importance of nodes in different neighborhoods. And Vazquez et al. [25] tried to fuse the MS-G3D algorithm with the 3D-CNN architecture S3D [26], which provided marginal improvements on the AUTSL dataset. Furthermore, Al et al. [18] also proposed a multi-head attention mechanism to enhance the spatial context representation based on S3D for SLR.

2.2. Graph convolution neural networks

Convolutional neural networks (CNNs) have achieved great success in processing image, video, audio and other gridded data. However, there are numerous applications involving data in non-Euclidean forms such as graphs. Using CNN to process graph data is not straightforward. How to operate on graph data has been extensively explored over the last few decades.

Graph convolution networks (GCNs) extended CNNs to handle graph-structure data with arbitrary shape. As such, it has a powerful representation capability for processing graph data [27,28]. Recently, various methods have been proposed to design the graph convolution [29–32], which mainly fall into two streams: spectral perspective and spatial perspective. The spectral perspective methods generated the graph representation in graph Fourier transform domain based on eigen-decomposition. For example, Defferrard et al. [31] designed fast localized convolutional filters by using recurrent Chebyshev polynomials. Kipf et al. [33] further simplified this approach using the first-order approximation of the spectral graph convolutions. However, these methods are limited by the computational efficiency due to the requirement of eigen-decomposition. By contrast, The spatial perspective methods directly performed convolution on the graph node and its neighbors. Niepert et al. [32] sampled the neighborhoods for each of the nodes based on their distances in the graph. To further improve the representation of GCN, graph attention network (GAT) [34] introduced the attention mechanisms to GCN in order to select critical information from all inputs. DimNet [35] introduced directed message passing to improve the capacity of graph representation.

Our work follows the spatial perspective methods. The implementation of graph convolution in ST-GCN [14] can be formulated as:

$$f_{\text{out}} = \sum_{k=1}^{K_v} W_k f_{\text{in}} A_k \odot M_k, \quad (1)$$

where K_v is the number of different partitions, W_k is the weight matrix for the k th partition to evolve the graph feature f_{in} , $A_k = A_k^{-\frac{1}{2}} \bar{A}_k A_k^{-\frac{1}{2}}$, where $\bar{A}_k$ is the specific adjacency matrix of the k th partition. A learnable edge importance mask M_k is inserted in the spatial graph convolution using the element-wise product operation (i.e., $\odot$), to make joints have different importance.

Shi et al. argued that it is not optimal for diverse samples by setting the graph topology manually and fixing it over all layers [36]. Thus, they proposed to divide the adjacency matrix into two parts: A_k and B_k .

$$f_{\text{out}} = \sum_{k=1}^{K_v} W_k f_{\text{in}} (A_k + B_k), \quad (2)$$

where A_k is an $N \times N$ adjacency matrix which represents the constant physical structure of the human body. B_k is an attention matrix, similar to mask M_k in equation (1). The specific initialization of B_k can improve the performance by enhancing the flexibility of the model.

3. Hand-aware graph convolution network

The input skeleton sequence $S \in \mathbb{R}^{C \times T \times N}$ represents temporal sequences of body joints in 2D/3D coordinates, where C represents the coordinate channels, T denotes the temporal length and N represents the number of joints. The skeleton sequence is processed into skeleton graph as input of network $G = (V, E)$, the node set $V = \{v_{i,t} | i = 1, \dots, N, t = 1, \dots, T\}$ includes all the joints in a skeleton sequence. The edge set E includes all intra-body skeletal connections in each frame and the inter-frame connections of the corresponding joints in adjacent frames.

The overall architecture of proposed method is shown in Fig. 2. First, the hand-aware graph convolution network layer is performed to extract features $F \in \mathbb{R}^{C_f \times T_f \times N}$, which focus on the hand topology information. Then, an adaptive DropGraph is introduced to remove the temporal and spatial redundancy in the skeleton sequence. Finally, the discriminative features extracted by these layer-stack network are fed into the softmax classifier for sign language recognition.

In this section, we first describe a hand-aware graph convolution network layer. Then an adaptive DropGraph is introduced. Finally, the constructed hand-aware graph convolution network and multi-stream approach are discussed.

3.1. Hand-aware graph convolution layer

As the most important local cue in the SLR task, the hand information can be considered as a prior [37]. However, the existing SLR solutions [38,39] include the global information of whole upper body skeleton structure, which did not strengthen the importance to local hand clue. To address this issue, we propose a hand-aware graph convolution layer which could simultaneously capture global

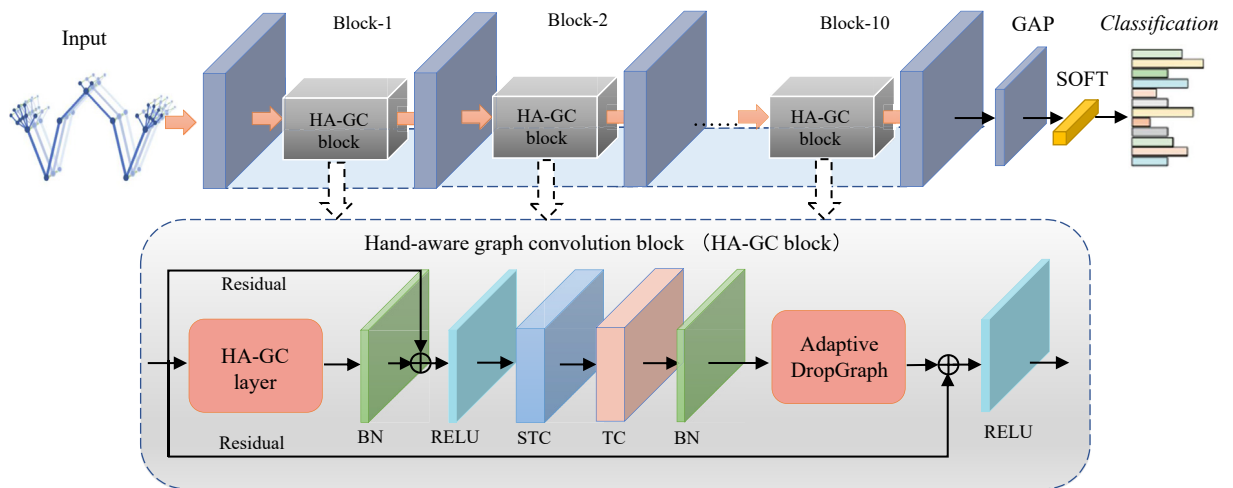


Fig. 2. Overview of our proposed hand aware graph convolutional network (HA-GCN) architecture, where the top is the pipeline of HA-GCN and the bottom illustrates the hand-aware graph convolutional (HA-GC) block. HA-GC block mainly include well-designed HA-GC layer and adaptive DropGraph module. Besides, it is also composed of batch normalization layer (BN), rectified linear unit (RELU), spatial temporal and channel (STC) attention, and temporal convolution layer (TC).

body and local hand information. Specifically, two hand-aware sub-graphs, named structured hand graph SH and parameterized hand graph PH respectively, are defined in the GCN layer to better model the hand topology structure.

3.1.1. Structured hand graph SH

In this section, we define a structured hand (SH) graph to represent the heuristic hand-region relationship to enhance the key local information. To be specific, we build the physical connection of hands and the direct connection of wrist as the SH graph. This graph can be formulated by the adjacent matrix SH as follows:

$$SH_{ij} = \begin{cases} 1, & d(v_i, v_j) = 1, \{i, j\} \in V_{\text{hand}} \\ 0, & \text{otherwise} \end{cases}, \quad (3)$$

where SH_{ij} indicates whether node i and node j are physically connected, $d(v_i, v_j) = 1$ indicates there exists a connection between node i and node j , V_{hand} denotes the set of the hand nodes.

3.1.2. Parameterized hand graph PH

We argue that the SH graph fixed by the physical connection is insufficient for some words in sign language. Therefore, we also construct a parameterized hand (PH) graph as a more flexible manner, which is represented by an adjacent matrix PH . Compared with SH graph, the elements of PH are parameterized and learned from the training data to fit the SLR task. It means that the elements in the matrix PH can be arbitrary values instead of just 0 or 1 in SH graph, which not only indicate the existence of connections between nodes, but also indicate the strength of the connection. On the one hand, the important nodes, such as the index finger and thumb, are paid more attention through learning the PH graph. On the other hand, the hand graph can be evolved through creating or deleting some edges during learning. This means that, in addition to physically adjacent nodes, distant nodes of the hand can also be directly connected in the constructed graph.

The constructed structured and parametric hand graphs are unique for different layers and samples, which greatly increases the flexibility of the model. After constructing these two hand graphs, a novel hand-aware graph convolution layer is designed to fuse body graphs (A, B) and hand graphs (SH, PH) to focus on the global information as well as local information. In addition, a gating mechanism is designed to adjust the weights between hand graphs SH and PH. The operation in the hand-aware graph convolution layer can be formulated as:

$$f_{\text{out}} = \sum_k^{K_v} W_k f_{\text{in}} (A_k + B_k + SH_k \cdot \alpha + PH_k \cdot \beta), \quad (4)$$

where A, B, SH and PH are different graphs introduced above. The α and β are parameterized coefficients which multiply with SH and PH respectively. These coefficients are unique to each layer and dynamically updated during training. K_v is the kernel size of convolution and W_k is the weighting function respectively, which are introduced in equation (1).

The detailed structure of the hand-aware graph convolution layer that implements equation (4) is shown in Fig. 3. As can be seen, a residual connection, similar to Ref. [40], is added for each layer, which allows the layer to be inserted into any existing models without

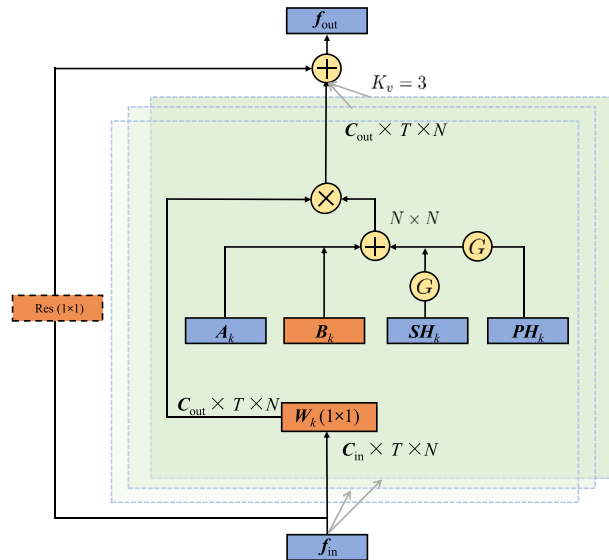


Fig. 3. Illustration of the hand-aware graph convolution layer. There are 4 types of graphs in each layer, i.e., A_k, B_k, SH_k and PH_k . The orange box indicates that the parameter is learnable. (1×1) denotes the kernel size of convolution. K_v denotes the kernel size of convolution. $\oplus$ denotes the element-wise summation. $\otimes$ denotes the matrix multiplication. The residual box (dotted line) is only needed when C_{in} is not the same as C_{out} .

breaking its initial behavior. Note that if the number of input channels is different from the number of output channels, a 1×1 convolution (orange box with dashed line in Fig. 3) is inserted in the residual path to transform the input to match the output in the channel dimension. By constructing and incorporating the structured hand (SH) graph and parameterized hand (PH) graph in the GCN layer, our proposed HA-GCN can better focus on hands trajectories in the SLR task compared with other GCN approaches that include the whole upper body joints without explicit focus on the hand topological structure.

3.2. Adaptive DropGraph

The spatial and temporal redundancy will inevitably exist during the sign language representation. It would no doubt result in the over-fitting problem and seriously interfere with the judgment of the sign label.

Therefore, it is necessary to drop the redundant frames and nodes. Furthermore, considering the inconsistency of redundancies between spatial and temporal dimensions, these two kinds of redundancies should be coped with individually. Inspired by DropGraph proposed by Ref. [16], we introduce an adaptive DropGraph module to avoid over-fitting in both spatial and temporal dimensions. In detail, a gating mechanism is designed to adjust the spatial and temporal DropGraphs to fit SLR tasks in a more flexible manner.

As shown in Fig. 4, we adopt a gating mechanism using the learnable coefficients δ and γ to optimize the spatial DropGraph and the temporal DropGraph, respectively. For the single DropGraph, the input feature $X \in \mathbb{R}^{N \times C \times V \times T}$ is down sampled by the mean operation, where N , C , V and T represent the batch size, number of channels, the number of joints and time sequence length respectively. Then, it is processed using Bernoulli generation to generate a mask $M \in \mathbb{R}^{N \times (C \times V) \times T}$. Finally, the obtained mask is multiplied with the input feature X to achieve the regularization.

3.3. Hand-aware graph convolution block and network

We construct a hand-aware graph convolution block using the above proposed hand-aware graph convolution layer and adaptive DropGraph strategy together with space, time and channel (STC) attention module [41] and temporal convolution (TC) layer as shown in Fig. 2. STC attention mechanism is composed of a spatial attention module, a temporal attention module and a channel attention module connected in cascade. Inspired by ST-GCN [14], TCN is adopted to handle the temporal features. To stabilize training and mitigate gradient explosion, a residual connection is also added in each base block.

For the construction of HA-GCN network, it is mainly composed of 10 hand-aware graph convolution blocks. The numbers of output channels for each block are 64, 64, 64, 64, 128, 128, 128, 256, 256 and 256. At the beginning, a batch normalization (BN) layer is added to normalize the input skeleton data. And finally, a global average pooling (GAP) layer is performed to pool the feature maps of different samples into the same size in the spatial and temporal dimensions. The result output by the fully connected (FC) layer is fed into the softmax classifier to obtain the prediction result.

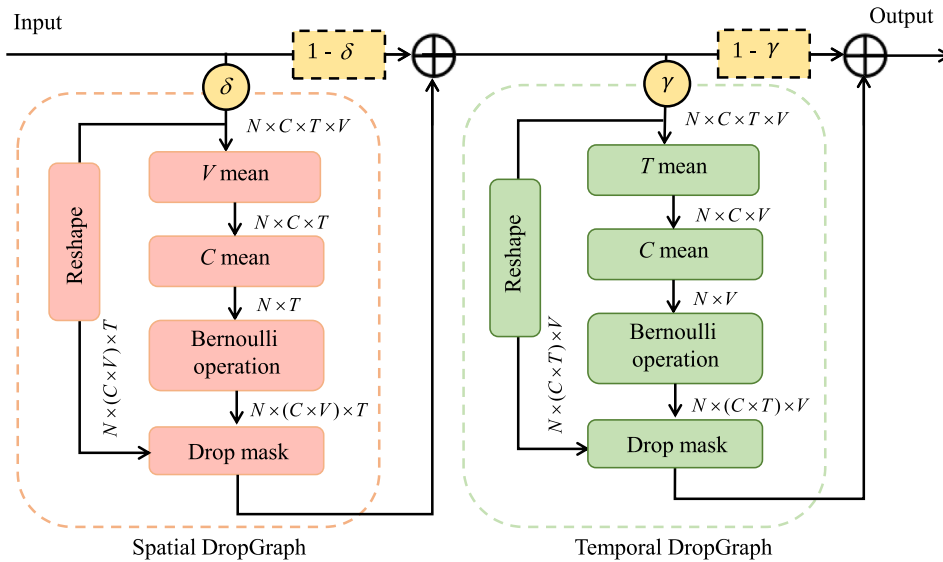


Fig. 4. Illustration of the adaptive DropGraph module, where spatial DropGraph and temporal DropGraph have a similar structure, which uses down sampling and Bernoulli generation successively to obtain the mask to enhance regularization effects on the input features. In addition, the above DropGraphs are concatenated using the gating mechanism.

3.4. Multi-stream approach

A multi-stream approach is implemented to further improve the recognition performance. In detail, both the first-order information (the coordinates of the joints) and the second-order information (the direction and length of the bones), as well as their motion information are proposed to be investigated for the skeleton-based SLR task. As such, joint data could be divided into joint stream (J-stream) and joint motion stream (J-M-stream), meanwhile bone data could be divided into bone stream (B-stream) and bone motion data stream (B-M-stream) [15,36,41]. Then these streams are fed into the above HA-GCN respectively.

The different streams have different characteristics and they are complementary to each other. In order to exploit the complementary information of these streams, three fusion strategies are designed to fuse multiple streams, including max fusion, average fusion, and adaptive fusion. These three fusion strategies all fuse the softmax scores of the four streams to predict the sign language label.

- Max fusion is designed to choose top-1 score vector as the predicted sign language label.
- Average fusion treats different streams equally and assigns the same fusion weight to each stream.
- Adaptive fusion uses the fully connected layer to learn the adaptive fusion weights.

4. Experiments

To verify the effectiveness of the proposed network on different scale datasets, we conduct experiments on two widely-used datasets, namely, the AUTSL and INCLUDE datasets. First, we describe the two datasets. Then, the experimental details of our experiments, including training setup and data pre-processing are described. Moreover, ablation studies are conducted on joint-stream data to verify the effectiveness of important components of hand-aware GCN. And multi-stream fusion are further analyzed. Finally, the performance comparisons in terms of accuracy performance and computation complexity between HA-GCN and other state-of-the-art methods are presented on both datasets.

4.1. Datasets

AUTSL [42] is a large-scale Turkish sign language dataset. Specifically, it contains 38,336 videos of 226 signs performed by 43 signers in 20 different backgrounds (indoor, outdoor settings and different lighting conditions.). The dataset is split into training (27,676 samples, 72%), validation (4884 samples, 13%) and testing subsets (5776, 15%). The AUTSL dataset provides RGB, depth, and skeleton modalities. The RGB-D data are captured by a Kinect V2 sensor [43,44] with a spatial resolution of 512×512 pixels and a time resolution of 30 FPS.

INCLUDE [45] is the first publicly available Indian sign language dataset, covering 263 sign language vocabularies recorded by 43 experienced signers. It contains 4287 sign language videos about 70,000 frames at a spatial resolution of 1920×1080 pixels. The video was shot in bright natural light, and the signer's costume or background was not deliberately regulated to provide a capture close to the real scene.

Table 1 summarizes the main characteristics contained of INCLUDE and AUTSL datasets. For above two datasets, each skeleton is composed of 133 junction points, whose location information are selected from HRNet pose estimation model [6] provided by MMPose [46]. The whole-body skeleton data is reduced and trimmed down from 133 nodes to 27 nodes. These 27 key points are considered as input to construct a topology graph, including 10 nodes of each hand and 7 nodes of the upper body. The detailed description about skeleton point ordinal number is shown in Fig. 5.

4.2. Experimental details

Our proposed HA-GCN is implemented on Pytorch, and NVIDIA TITAN XP GPU is used to train the network. Since the AUTSL and INCLUDE datasets used in this paper will be processed through the same pre-processing steps, the hyper parameter settings of networks for different datasets would be also consistent.

4.2.1. Training setup

There is no need to use the pre-training model to initialize the parameters at the early stage of training. Cross entropy loss is adopted as the loss function. The coefficients α , β , δ and γ , are randomly initialized and dynamically updated during training. The SGD optimizer

Table 1
A statistical summary of AUTSL and INCLUDE datasets.

Property	AUTSL	INCLUDE
Language	Turkish	Indian
Number of signs	226	263
Number of signers	43	7
Total samples	36,302	4287
Average samples per sign	169.6	16.3
Train samples	27,676	3468
Validation samples	4884	–
Test samples	5776	816

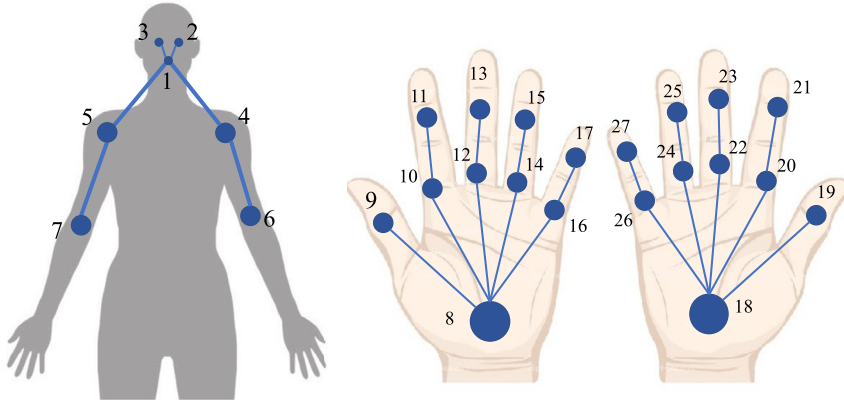


Fig. 5. Skeleton point ordinal number for the AUTSL and INCLUDE datasets

is used with the weight decays as $1e^{-4}$. The batch size is set to 64. The training iterations number is set to 500 epochs, and when the accuracy of validation set fails to increase after 50 epochs, the early stop strategy is performed immediately to avoid excessive meaningless training. The learning rate adopts a warm-up strategy. Specifically, the learning rate is increased from 0.005 to 0.1 in the first 20 epochs, then maintain it at 0.1 until the 150th epoch, and keep the learning rate at 0.01 from the 150th epoch to the 200th epoch, finally keep training with a learning rate of $1e^{-3}$ from the 200th epoch.

4.2.2. Data preprocessing

For multi-stream framework, the skeleton data is firstly processed into multiple data streams such as joints, bones, joint motion and bone motion. And the crop length of samples is set to 150. If a video has less than 150 frames, we repeats the video until it reaches 150 frames.

During the training, for each video sample, we adopts the following data enhancement steps: a) Random sampling. Randomly select the starting position of the video and take a clip of length 100 frames. Thus, The input sample is resized to $3 \times 100 \times 27$. b) Random shift. For the joint stream, the random shift is performed on all coordinates one by one, i.e. displacement operation. For bone, joint motion and bone motion streams, random shift is only performed on the origin coordinate (coordinate of nose). c) Random mirror. A random number determines whether to perform a mirroring operation. This random number has the value range $[0,1]$, and the mirroring operation is performed when the random number is greater than 0.5. Specifically, the mirroring operation is to make the data coordinate values centrosymmetric along the central axis of the video frame and to swap the order of the symmetric joints. 4) Normalization. The coordinates of the above nodes are normalized to $[-1, 1]$. For joint stream, subtract the mean value of the origin coordinate for all joints one by one. For other streams, only the origin of coordinates are normalized.

4.3. Ablation study

4.3.1. Hand-aware graph convolution layer

As introduced above, the two different sub-graphs, structured hand (SH) and parameterized hand (PH) graphs, are introduced in the GCN layer to focus on the hand graph. Specifically, SH graph is designed to build the natural connection of hands to represent the heuristic hand-region relationship and enhance the key local information. PH graph is parameterized and made to learn from the training data to draw non-physical dependencies among joints for various actions. Besides, a gating mechanism is adopted for adaptive learning involving two learnable coefficients α and β . In the experiments, the effectiveness of the combination of these components is verified, including PH graph, SH graph and gating mechanism, by comparing with the original ST-GCN. Table 2 shows the top-1 recognition accuracy on the AUTSL and INCLUDE datasets.

It can be seen from Table 2 that (1) either the proposed sub-graphs SH or PH are beneficial for the SLR task with an accuracy improvement of about 3% and 0.24% on the AUTSL and INCLUDE datasets. That is because that the two different sub-graphs, structured hand (SH) and parameterized hand (PH) graphs, introduced in the GCN layer enable the network to focus on the hand representation. Specifically, the SH graph build the natural connection of hands to represent the heuristic hand-region relationship and enhance the key local information. PH graph parameterized and learned from the training data to learn non-physical dependencies among joints for various actions. (2) When these two sub-graphs are fused together by summation without gating (as the fourth row shows), the model achieves significant performance improvement by 0.11% at recognition accuracy on the AUTSL dataset, compared with that when SH or PH applied individually. The experimental results demonstrate the complementation between SH and PH. It is worth noting that, for the INCLUDE dataset, the accuracy decrease mainly because the INCLUDE dataset is relative small. Limited by the tiny scale of dataset, the fuse of two sub-graphs makes the network hard to train due to overfitting. (3) It clearly reflects from the results that the gating mechanism further optimizes relationship between SH and PH during the training. The recognition accuracy is improved by about 0.6% compared with fusion by summation. And the accuracy reaches as high as 95.88% and 99.39% on the AUTSL and INCLUDE datasets.

Table 2

Comparison of HA-GCN variants at recognition accuracy on the test set. The first row corresponds the baseline ST-GCN without SH or PH.

SH	PH	Gating	Top-1 (%)	
			AUTSL	INCLUDE
–	–	–	91.99	99.02
✓	–	✓	95.06	99.26
–	✓	✓	95.24	99.26
✓	✓	–	95.35	98.90
✓	✓	✓	95.88	99.39

The best result in each column is in bold.

4.3.2. Key modules in HA-GCN block

Ablation studies are performed on the proposed hand-aware graph convolution (HA-GCN) block to verify the effectiveness of each component. The proposed HA-GCN block is mainly composed of hand-aware graph convolution layer, STC attention module and the adaptive DropGraph. Hand-aware graph convolution layer introduce two different sub-graphs, structured hand (SH) and parameterized hand (PH) graphs. Besides, the STC module and the adaptive DropGraph module are also important modules introduced in HA-GCN block, as shown in Fig. 6. In Table 3, we manually remove each crucial component in the hand-aware graph convolution block to verify the effectiveness. The ablation experiments are performed on the AUTSL and INCLUDE dataset at the top-1 recognition accuracy.

By comparison, we can observe that without the SH and PH graphs in SLR-GCN block, the recognition accuracy is reduced with a large margin of 3.89% under the same experimental conditions. It indicates that capturing the hand topology information is crucial to the SLR task. Besides, by removing the adaptive DropGraph module, the performance also decreased by nearly 3.2%, which indicates that the adaptive DropGraph both in spatial and temporal dimensions can learn more suitable features for SLR task. In addition, with the STC attention module, our network achieves better accuracy by 0.56% and 0.37% on the AUTSL and INCLUDE datasets, respectively.

4.3.3. Multi-stream fusion

Table 4 compares the performance of using different kinds of input data (shown as joint stream, bone stream, joint motion information and bone motion stream as described above), respectively.

The results show that, for single-stream methods, joint streams slightly outperform the bone streams among the four data streams, which reveals that the joint stream contains more useful patterns contributing to the final results. Besides, motion streams (joint motion and bone motion) are generally of lower performance than the joint and bone stream. However, adding them together is going to lead to improvement. Finally, compared with that of single data stream, the integrating of multi-stream skeleton data stably improved the SLR

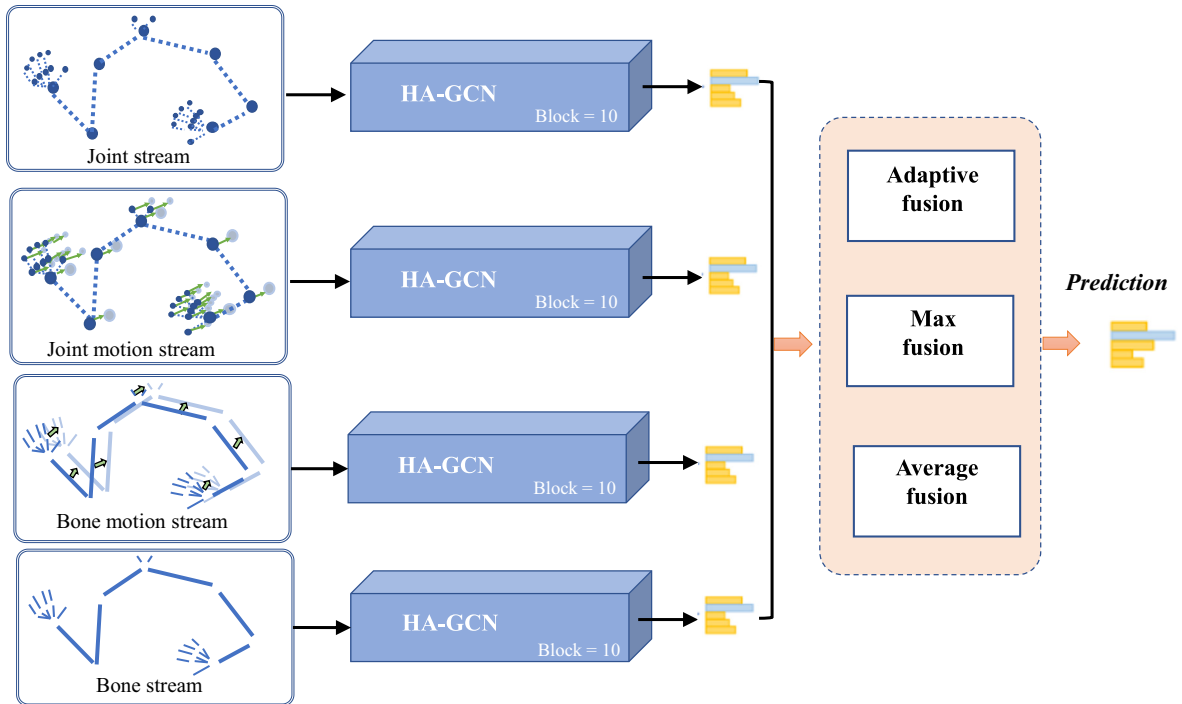


Fig. 6. Illustration of the overall architecture of the multi-stream HA-GCN. For each HA-GCN, it composed of 10 HA-GC blocks. The softmax scores of the four streams are fused using different fusion strategies, including max fusion, average fusion, and adaptive fusion, to obtain the final prediction.

Table 3

Ablation results for components in HA-GCN on AUTSL and INCLUDE datasets.

Variation	Top-1 (%)	
	AUTSL	INCLUDE
Our HA-GCN	95.88	99.39
Wo SH & PH	91.99	99.02
Wo adaptive DropGraph	92.68	98.90
Wo STC module	95.32	99.02

The best result in each column is in bold.

Table 4

Results of multi-stream fusion based on HA-GCN.

Data stream branch	Top-1 (%)	
	AUTSL	INCLUDE
Joint stream	95.88	99.39
Bone stream	95.11	99.26
Joint motion stream	93.91	96.94
Bone motion stream	93.37	97.30
Multi-stream fusion	96.82	99.63

The best result in each column is in bold.

performance. This phenomenon verifies the information complementation between different streams. The additional motion modality and bone features improve the discrimination.

4.3.4. Effects of fusion schemes

In this subsection, the performance of different fusion schemes are evaluated in Table 5, including max fusion, average fusion and adaptive fusion. In the adaptive fusion, we initialize the network with the parameters trained from average fusion, then freeze all the layers except the fusion layer. As shown in Table 5, we can observe that the average fusion brings about 0.65% and 0.12% improvement (AUTSL and INCLUDE dataset, respectively) compared with max fusion. Average fusion could incorporate the complementary and versatile features from different streams and achieves better performance. Beside, among these methods, the adaptive fusion achieves the best performance and brings about 0.13% and 0.12% improvement (AUTSL and INCLUDE dataset, respectively) compared with average fusion, since the network could learn the combination of high-level representations adaptively. Moreover, we observe that joint modality is assigned a highest weight when allowing the network to fusion adaptively. This result suggests that the network is most confident with the scores from joint stream. It is probably because that joint modality is the intrinsic high-level representation of human body, which promote to capture more useful patterns.

4.4. Comparisons with the state-of-the-art methods

Table 6 shows the comparison results between our HA-GCN and the state-of-the-art skeleton-based SLR methods in terms of accuracy on both AUTSL and INCLUDE datasets. The methods used for comparisons are divided into LSTM-based methods, transformer-based methods and GCN-based methods. As can be seen from Table 6, our proposed model outperforms state-of-the-art methods with a great margin on the AUTSL and INCLUDE datasets, which verifies the excellence of our model. To be specific, 1) LSTM-based methods [42,45] and transformer-based methods [47] treat skeleton data as input vector, which neglect the spatial structure information of the skeleton. 2) GCN-based methods have achieved promising progress on the SLR task, all of which achieve the accuracy rates superior to 90%. That makes GCN based methods become mainstream approach of skeleton-based SLR task. ST-GCN [15] is the first network that uses spatio-temporal graph convolution. Based on ST-GCN, other GCN based approaches, including SL-GCN [15],

Table 5

Top-1 accuracy (%) of different fusion schemes with different weights based on HA-GCN. The weights represent joint, bone, joint motion, bone motion in turn.

Ensemble	AUTSL		INCLUDE	
	Weights	Top-1 (%)	Weights	Top-1 (%)
Max	–	96.04	–	99.39
Avg	[1.0, 1.0, 1.0, 1.0]	96.69	[1.0, 1.0, 1.0, 1.0]	99.51
Adaptive	[1.69, 1.68, 1.05, 1.54]	96.82	[1.48, 1.44, 0.76, 0.35]	99.63

Table 6

Comparison of the propose HA-GCN with the state-of-the-art skeleton based SLR networks.

Network	FLOPs	Params	Top-1 (%)	
			AUTSL	INCLUDE
CNN + FPM + Bi-LSTM + Attention [42]	–	–	62.02	–
MobileNetV2 + Bi-LSTM [45]	3.77 M	2.85 G	–	85.66
LSTM [47]	1.65 M	0.15 G	77.45	82.97
Transformer [47]	4.33 M	0.43 G	81.03	90.44
ST-GCN [14]	3.12 M	2.94 G	90.41	91.30
SL-GCN [15]	2.83 M	2.32 G	95.02	95.59
TL-GCN [48]	3.17 M	3.01 G	95.46	98.90
AAGCN [41]	3.81 M	3.36 G	94.90	98.41
Our HA-GCN	3.48 M	3.01 G	95.88	99.39
Our multi-stream HA-GCN	3.48 M ^{×4}	3.01 G ^{×4}	96.82	99.63

The best result in each column is in bold.

TL-GCN [48] and AAGCN [41], have made some improvements to further boost the performance. By comparison, our HA-GCN has the largest performance improvement without significantly increasing the computational and parameter costs, 5.47% and 8.09% over ST-GCN on the AUTSL and INCLUDE datasets, respectively. In addition, our multi-stream HA-GCN further improves the accuracy performance by 0.84% and 0.24% on the two datasets by exploiting the complementary information of multi-streams.

4.5. Model complexity

To demonstrate the practicality of our proposed method, we conducted a comparative analysis of model complexity in terms of the number of model parameters (Params), floating-point operations (FLOPs), with the results presented in Table 6. Due to the lack of reported FLOPs and parameter numbers for most models, we obtain the results by their released codes. As can be seen, our proposed method manifests advantage both in model efficiency and accuracy. Specifically, compared with other GCN based SLR approaches including ST-GCN, SL-GCN and TL-GCN, our method achieves significantly better performance with comparable computation complexity. Compared with AAGCN, the proposed multi-stream HA-GCN achieve performance gains with parameter reduction.

5. Visualization and discussion

In this section, to illustrate that our HA-GCN can capture rich feature representations by using dynamic hand topologies, we visualize some representations of HA-GCN, including the detailed comparison on certain class accuracy, the learned joint-scale hand-aware graphs.

5.1. Detailed comparison on certain class accuracy

In Fig. 7, we compare the recognition accuracy between our HA-GCN and different GCN-based methods on 10 most confusing sign language words (i.e., Minute, Request, They, Tree, Who and Bill, Brother, Cold, Red, Television.) from AUTSL and INCLUDE datasets. We can observe that the proposed HA-GCN performs best on the most confusing sign language words compared with other GCN-based methods. Surprisingly, the recognition accuracy of most confusing sign language words can be improved from less than 70% up to 100% by our proposed HA-GCN such as “Tree” words in AUTSL and “Television” in INCLUDE. This experimental results validates the effectiveness of HA-GCN for SLR.

5.2. The learned joint-scale hand-aware graphs

It is the commonsense that the interaction between hands plays an important role in the SLR task. To demonstrate the capability of the proposed HA-GCN for capturing the interaction between hands in each layer, we visualize the connections between joints on the skeleton sketches from block 1 to 10. As shown in Fig. 8, SL-GCN heuristically predefine the natural-connected joints as graph-structure, which merely represents the physical structure and does not focus on the connection between hands which is very important for SLR tasks. Though the TL-GCN, R-GCN and MS-AAGCN could adaptive learn the graph structure designed for SLR, the visualization show that these methods are able to construct structure-learnable graph in only several blocks. It can be seen from Fig. 8 that our proposed HA-GCN adaptively learns the complex hand topological relationships beyond the physical connection. Moreover, our proposed HA-GCN adaptively learns the hand topology in earlier blocks. This visualization results confirm the effectiveness and necessity of the proposed HA-GCN in learning hand topology information for the SLR task.

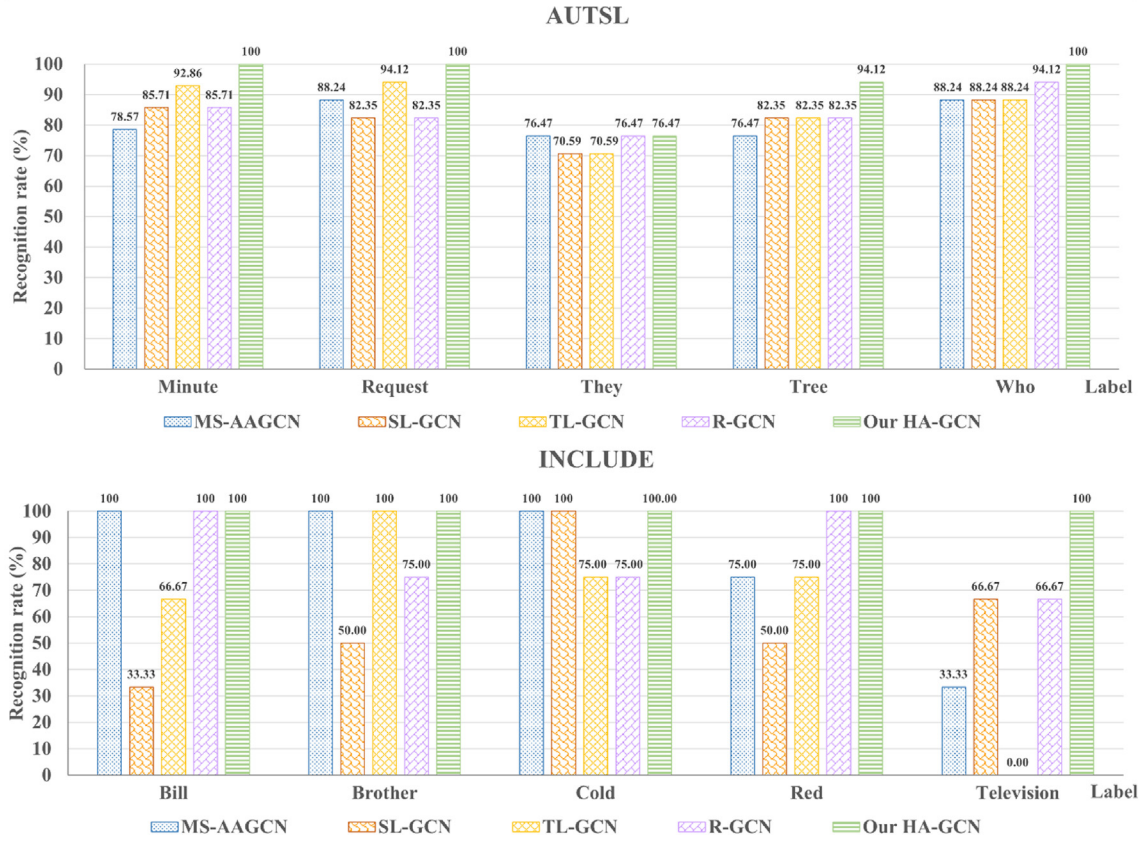


Fig. 7. Accuracy comparisons of the different GCN-based methods on 5 most confusing sign language words from AUTSL (top) and INCLUDE (bottom).

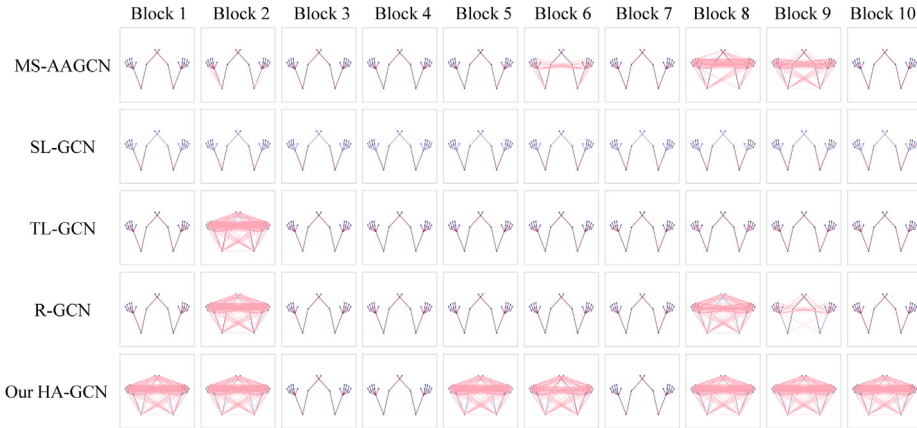


Fig. 8. Comparison of adjacency matrix visualization of graph convolution networks. The blue lines show the upper body physical connections. The thickness of red lines shows the connection strength of the learned adjacent matrices.

6. Conclusion

In this paper, we propose a novel hand-aware graph convolution network (HA-GCN) to focus on hand topological relationships for skeleton-based SLR. Specifically, two hand graphs are defined and incorporated to focus on the hand topology information. Based on these two sub-graphs, a hand-aware graph convolution layer is designed to capture multi-scale features including both global body and local hand information. In addition, an adaptive DropGraph is proposed in hand-aware graph convolution block to remove the spatial and temporal redundancy in the sign language representation. And to further improve the performance, the information of joint data,

bone data, joint motion data and bone motion data are simultaneously modeled in a multi-stream framework. The final experimental results show that the proposed multi-stream HA-GCN is beneficial to skeleton topology data to achieve the best recognition performance. In the future works, we will extend to focus on the fusion of multimodalities, such as RGBs, depth images and so on, for SLR without sacrificing efficiency.

CRedit authorship contribution statement

Juan Song: Methodology. **Huixuechun Wang:** Software. **Jianan Li:** Methodology. **Jian Zheng:** Software. **Zhifu Zhao:** Writing – review & editing. **Qingshan Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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